

PROJECT ARCHITECTURE – AI WeatherWise API

1. Architecture Overview

The AI WeatherWise API follows a microservices-based cloud architecture that integrates weather data providers, AI analytics, and client applications.

2. Architectural Components

Client Layer: Web, Mobile, IoT, and third-party integrations using React, Angular, Flutter, Android and iOS.

API Gateway: Authentication, API key validation, rate limiting, logging, and request routing.

Weather Service: Current weather, forecasts, and air quality reporting.

Alert Service: Storm alerts, flood warnings, heatwave notifications.

AI Insight Service: Travel, agriculture, outdoor activity, and energy optimization recommendations.

3. Data Layer

Weather Database stores forecasts, alerts, weather records, and user preferences. Redis is used for caching.

4. AI & Analytics Layer

Pipeline: Data Cleaning -> Feature Engineering -> ML Training -> Prediction Models -> AI Recommendations. Models include Random Forest, XGBoost, and LSTM.

5. Security Architecture

OAuth 2.0, JWT authentication, HTTPS/TLS encryption, rate limiting, and data encryption at rest.

6. Deployment Architecture

Load balancer, application servers, Redis cache, PostgreSQL database, and AI processing node deployed on Azure, AWS, or GCP with Docker and Kubernetes.

7. System Workflow

User request -> API Gateway -> Weather Service -> Data Processing -> AI Analysis -> Recommendation Response.

8. Architecture Benefits

Scalability, high availability, fast response time, AI-powered insights, secure access, and real-time weather intelligence.